

Certificate-Only Causal Discovery: Anytime-Valid Adjacency and Orientation

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Abstract

Constraint-based causal discovery removes an edge when a conditional-independence test *fails to reject*, converting absence of evidence into evidence of absence. This is why finite-sample correctness for such methods requires faithfulness, and it is why they cannot say which of their outputs the data actually supports. We propose a discovery algorithm built on a single rule: assert a feature of the graph only by *rejecting* a null that is true whenever that feature is absent, and never assert anything by failing to reject. Two certificates implement the rule. For adjacency, the null “some conditioning set separates the pair” is a union null, and the minimum of the per-set e-processes is an e-process for it; crossing certifies an edge, using no faithfulness assumption. For orientation, the two directions of a pair induce different joint densities under a linear non-Gaussian model, so a sequential universal-inference e-process against the reversed factorisation certifies an arrowhead. Both are anytime-valid, so the estimate may be monitored continuously and the run stopped on any data-dependent rule. The resulting algorithm grows from the empty graph, reports undecided pairs as undecided, and bounds the probability that any certified edge or arrowhead is ever wrong, uniformly over stopping times. Empirically, on instances violating δ -strong faithfulness the algorithm’s error stays within budget while delete-on-non-rejection methods fail in 60–80% of runs; and 96% of true edges are oriented against a Markov-equivalence-class ceiling of 33%, because orientation error control and orientation beyond the equivalence class have previously been available only separately.

Keywords: causal discovery; e-values; anytime-valid inference; safe testing; family-wise error rate; LiNGAM

1 Introduction

A causal discovery algorithm returns a graph, and a practitioner reading that graph would like to know which of its features the data support. Constraint-based methods—PC, FCI and their descendants [Spirites et al., 2000]—answer this question poorly, for a structural reason. They begin from the complete graph and delete an edge whenever a conditional-independence test fails to reject. Failure to reject is not evidence of independence: it is compatible with independence, with a weak dependence, and with a test that lacked power. The deletion nevertheless happens, and the resulting absence of an edge is reported with the same authority as its presence.

Two consequences follow. First, finite-sample guarantees for these methods require faithfulness, or a quantitative strengthening of it; the assumption is doing the work that the data cannot. Second, the output cannot distinguish “this edge is absent” from “nothing was learned about this pair”.

This paper takes the opposite discipline. We adopt a single rule:

Assert a feature of the graph only by rejecting a null hypothesis that is true whenever that feature is absent. Never assert anything by failing to reject.

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The rule is restrictive, and its consequences are not cosmetic. An algorithm obeying it cannot begin from the complete graph, because it has no licence to delete; it begins from the empty graph and adds. It cannot report an edge as absent, only as undecided. And what it controls is the probability of asserting something false, rather than the probability of missing something true.

Making the rule operational requires two certificates, and the second has no direct precedent.

Adjacency. For a pair (i, j) the natural null is composite: *some* conditioning set separates them. This is true whenever the pair is non-adjacent, so rejecting it certifies an edge. Because the null is a union, the minimum of the per-set e-processes is an e-process for it (Section 3). Two things follow that matter more than the construction. Validity uses no faithfulness assumption—if i and j are non-adjacent then $\text{pa}(i)$ or $\text{pa}(j)$ separates them, so the null holds—and multiplicity runs over pairs rather than over (pair, conditioning set) triples.

Orientation. Orientation error is not uncontrolled in the literature: PC-p [Strobl et al., 2017] tests unshielded colliders and Meek-rule orientations and controls their false discovery rate. But those tests are confined to the Markov equivalence class, and an edge lying in no v-structure is unorientable in principle by any of them. Conversely the functional-model methods that escape the equivalence class—LiNGAM and its descendants [Shimizu et al., 2006, 2011]—choose a direction by comparing scores, with no error control on the choice. Under a linear non-Gaussian model the two orientations of a pair induce different joint densities and exactly one is correct, so “the direction is $j \rightarrow i$ ” is a genuine null; a sequential universal-inference e-process against it certifies the arrowhead $i \rightarrow j$ (Section 4). Under Gaussian noise the two factorisations are observationally indistinguishable, the process does not grow, and no certificate is issued—the procedure abstains exactly where the direction is unidentified.

Contributions.

1. An e-process for the composite “separability” null obtained as a minimum over conditioning sets, giving time-uniform certification of adjacency under the Markov condition without faithfulness (Section 3).
2. A sequential certificate for causal *orientation* that is valid at any stopping time and is not confined to the Markov equivalence class, abstaining automatically under Gaussianity (Section 4).
3. CERT-CD, an algorithm combining both under one budget, whose output is a three-valued graph in which every asserted edge and arrowhead carries a certificate and undecided pairs are explicit, with a whole-graph guarantee that is proved rather than assumed (Section 5).
4. An empirical study separating instances by whether the premise of the competing guarantee holds, which shows that near-unfaithful instances cost the method power rather than validity (Section 7).

We are explicit throughout about what is not new. Section 6 gives that accounting in detail.

2 Background

2.1 E-processes and Ville’s inequality

An *e-variable* for a null H_0 is a non-negative statistic with $\mathbb{E}_P[E] \leq 1$ for all $P \in H_0$. An *e-process* is a non-negative process $(E_t)_{t \geq 0}$ such that $\mathbb{E}_P[E_\tau] \leq 1$ for every stopping time τ and

every $P \in H_0$ [Ramdas et al., 2023, Grünwald et al., 2024]. Ville’s inequality then gives

$$P\left(\exists t \geq 0 : E_t \geq 1/\alpha\right) \leq \alpha, \quad (1)$$

a bound holding *uniformly in time*. This is what licenses continuous monitoring: the analyst may inspect the evidence after every observation and stop whenever they wish, without inflating the error rate. A fixed-sample p-value offers no such licence, and the gap is large in practice (Section 7.1).

2.2 Causal discovery setting

Let $\mathcal{G}^*$ be a directed acyclic graph over variables $X_1, \dots, X_d$ and let P be Markov to $\mathcal{G}^*$. Write $\mathcal{S}_k(i, j)$ for the set of all conditioning sets $S \subseteq V \setminus \{i, j\}$ with $|S| \leq k$; this family is fixed before any data are seen. We say a pair is *separable within $\mathcal{S}_k$* if $X_i \perp\!\!\!\perp X_j \mid X_S$ for some $S \in \mathcal{S}_k$.

Assumption 1 (Separator size). Every non-adjacent pair of $\mathcal{G}^*$ has a separating set of size at most k .

Assumption 1 is structural rather than distributional and is satisfied whenever k is at least the maximum in-degree, since $\text{pa}(i)$ d-separates i from any non-adjacent j that is not a descendant of i , and $\text{pa}(j)$ does so otherwise.

3 The adjacency certificate

Fix a pair (i, j) and consider the composite null

$$H_{ij}^{\text{sep}} : \exists S \in \mathcal{S}_k(i, j) \text{ such that } X_i \perp\!\!\!\perp X_j \mid X_S. \quad (2)$$

By the Markov property and Assumption 1, H_{ij}^{sep} holds whenever i and j are non-adjacent in $\mathcal{G}^*$. Rejecting it therefore certifies an edge. Let $(E_t^{(S)})$ be any e-process for the point null $X_i \perp\!\!\!\perp X_j \mid X_S$, and define

$$A_t(i, j) = \min_{S \in \mathcal{S}_k(i, j)} E_t^{(S)}(i, j). \quad (3)$$

Proposition 2. (A_t) is an e-process for H_{ij}^{sep} .

Proof. Let $P \in H_{ij}^{\text{sep}}$, so $X_i \perp\!\!\!\perp X_j \mid X_{S^*}$ under P for some $S^* \in \mathcal{S}_k(i, j)$. Then $A_t \leq E_t^{(S^*)}$ pointwise, and $E^{(S^*)}$ is an e-process for a null that P satisfies, so for every stopping time τ , $\mathbb{E}_P[A_\tau] \leq \mathbb{E}_P[E_\tau^{(S^*)}] \leq 1$. $\square$

That a minimum of e-values is an e-value for a union null is standard [Vovk and Wang, 2021]; what matters here is the identification of adjacency as the complement of such a union, and the consequences below.

Remark 3 (No faithfulness). The proof uses only that the null (2) is true for non-adjacent pairs, which follows from the Markov condition and Assumption 1. Faithfulness is needed only for *power*: specifically adjacency-faithfulness [Ramsey et al., 2006], which is what makes every $E^{(S)}$ grow for a genuinely adjacent pair. An instance where it fails leaves pairs undecided rather than producing wrong edges. Stated exactly, validity requires three things—the Markov condition, Assumption 1, and a valid per-set e-process. The third is not free: our default primitive is a linear-Gaussian safe test (Section 3.1), so that model is assumed there, and substituting a nonparametric sequential test [He and Sutherland, 2026] removes it. Proposition 2 is indifferent to which primitive is used.

Remark 4 (Multiplicity over pairs). One hypothesis is tested per pair, so a union bound runs over $\binom{d}{2}$ hypotheses rather than $\binom{d}{2} \cdot |\mathcal{S}_k|$. At $d = 6$, $k = 2$ this is a $24\times$ larger per-hypothesis level than a triple-indexed budget.

3.1 The per-set primitive

For linear-Gaussian data we use an exact e-process obtained from a Bayes factor with right-Haar nuisance priors. Writing the hypothesis in regression form $X_i = X_S^\top \alpha + \beta X_j + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, the null is $\beta = 0$ with nuisance (α, σ) . Placing $\pi(\alpha, \sigma) \propto \sigma^{-1}$ on the nuisance block under both hypotheses and $\beta \mid \sigma \sim N(0, \sigma^2 g)$ on the tested coefficient, and integrating in closed form, gives

$$E_n = (1 + G)^{-1/2} (1 - \kappa \hat{r}_n^2)^{-(n-p)/2}, \quad G = g S_{yy}, \quad \kappa = \frac{G}{1+G}, \quad (4)$$

where $p = |S| + 1$, S_{xx}, S_{xy}, S_{yy} are cross-products of the X_S -residualised vectors and $\hat{r}_n^2 = S_{xy}^2 / (S_{xx} S_{yy})$ is the squared sample partial correlation. Two properties drive the implementation. First, E_n depends on the data only through Schur complements of the running Gram matrix, so a single $O(d^2)$ sufficient statistic serves the entire hypothesis family and no per-hypothesis state is stored. Second, g must be fixed in advance—a prior scale adapting to the running sample size destroys the martingale property—so we set it from a warm-up prefix excluded from every e-process.

Calibration was verified by simulation rather than asserted: monitoring after every observation over 3000 replications and 1500 time points gave realised rates of 0.149, 0.076, 0.041 and 0.008 against nominal $\alpha \in \{0.20, 0.10, 0.05, 0.01\}$.

Existing sequential conditional-independence tests largely operate under the Model-X framework, which requires the conditional law of X_i given X_S ; this is typically inaccessible in discovery, an obstacle also noted by Csillag et al. [2025]. Construction (4) avoids it.

4 The direction certificate

Under a linear non-Gaussian acyclic model [Shimizu et al., 2006] the two orientations of a pair induce different joint densities and exactly one is correct. Hence

$$H_{j \rightarrow i}^{\text{dir}} : \quad \text{the pair was generated by the } j \rightarrow i \text{ factorisation} \quad (5)$$

is a null that is true whenever the arrowhead $i \rightarrow j$ is absent, and rejecting it certifies that arrowhead. Conditioning on a covariate block Z , the two competing factorisations are

$$D^{j \rightarrow i} : \quad X_j = a^\top Z + u, \quad X_i = \beta X_j + \gamma^\top Z + e, \quad (6)$$

$$D^{i \rightarrow j} : \quad X_i = b^\top Z + v, \quad X_j = \delta X_i + \eta^\top Z + w, \quad (7)$$

both being conditional densities of (X_i, X_j) given Z , so the marginal law of Z cancels. We test by sequential universal inference [Wasserman et al., 2020]:

$$\log E_t = \sum_{s \leq t} \log D_{\hat{\theta}_{s-1}}^{i \rightarrow j}(o_s) - \sup_{\theta} \sum_{s \leq t} \log D_{\theta}^{j \rightarrow i}(o_s), \quad (8)$$

with the numerator's parameters fitted on strictly past data.

Proposition 5. *(E_t) defined by (8) is an e-process for $H_{j \rightarrow i}^{\text{dir}}$.*

Proof. Fix the true null parameter θ^* and set $M_t = \prod_{s \leq t} D_{\hat{\theta}_{s-1}}^{i \rightarrow j}(o_s) / D_{\theta^*}^{j \rightarrow i}(o_s)$. Since $\hat{\theta}_{s-1}$ is $\mathcal{F}_{s-1}$ -measurable and $D_{\hat{\theta}_{s-1}}^{i \rightarrow j}$ is a probability density, $\mathbb{E}[M_s \mid \mathcal{F}_{s-1}] = M_{s-1}$, so M is a non-negative martingale with $M_0 = 1$. The denominator of (8) is a supremum over the null model and therefore at least the likelihood at θ^* , giving $E_t \leq M_t$ pointwise. Optional stopping yields $\mathbb{E}[E_\tau] \leq \mathbb{E}[M_\tau] \leq 1$. $\square$

We fit each factorisation with generalised Gaussian noise, whose shape parameter spans Laplace through Gaussian to near-uniform, by profiling the shape over a grid with iteratively reweighted least squares for the coefficients.

Remark 6 (Abstention under Gaussianity). When the noise is Gaussian the two factorisations are observationally indistinguishable, the ratio (8) does not grow, and no certificate is issued. The procedure abstains precisely where the direction is not identified rather than guessing. Ding and Zhang [2025] give the underlying equivalence: Gaussianity of the forward-model noise is equivalent to independence between regressor and residual in the reverse regression.

Remark 7 (Two implementation constraints). Both were errors before they were understood. First, the observations scored by the numerator and those entering the denominator’s maximised likelihood must be the *same* index set; a denominator ranging over additional observations subtracts their log-density with no matching numerator term, adding a constant of order (extra observations) to $\log E_t$, which in our testing certified *both* directions of every pair, including under Gaussian noise. Second, the conditioning block Z is frozen when a pair’s adjacency is certified and the direction process accumulates only from that point; a block that kept changing would change the hypothesis under test, whereas (1) bounds a fixed one.

Remark 8 (Misspecification). Proposition 5 needs the denominator’s supremum to dominate the likelihood at the true null parameter, which is not guaranteed when the noise lies outside the fitted family. This is the certificate’s main theoretical risk. Empirically it is more robust than that suggests: against noise breaking each family property in turn—Student- t_3 (polynomial rather than exponential tails), shifted exponential (asymmetric) and a two-component mixture (bimodal)—the reversed direction was certified in 0 of 12 runs in each case at $\alpha = 0.05$, with full power retained.

5 CERT-CD

The algorithm maintains a three-valued adjacency status for every pair. It streams data in batches; after each batch it updates the running maximum of $A_t(i, j)$ for every undecided pair and certifies adjacency when the maximum crosses its threshold, at which point the two direction processes for that pair are opened with a frozen conditioning block. The output graph contains only certified edges; an edge carries an arrowhead when a direction certificate has fired, and circles at both ends otherwise. **An absent edge means undecided, not certified absent.**

Budgets are fixed before any data arrive: $\alpha = \alpha_A + \alpha_D$, with α_A split evenly over the $\binom{d}{2}$ adjacency nulls and α_D over the $d(d-1)$ direction nulls.

Theorem 9 (Whole-graph anytime validity). *Let τ be any stopping time, including one chosen by inspecting the output. Under the Markov condition, Assumption 1, valid per-set e-processes, and—for the direction certificates—the linear non-Gaussian model of Section 4,*

$$P\left(\exists \tau : \text{some certified edge is absent from } \mathcal{G}^* \text{ or some certified arrowhead is wrong}\right) \leq \alpha. \quad (9)$$

Proof. Each certificate is issued when an e-process for a null that holds whenever the asserted feature is absent crosses $1/(\alpha_{\bullet} w_{\bullet})$. By Proposition 2, Proposition 5 and Ville’s inequality (1), each is falsely issued with probability at most $\alpha_{\bullet} w_{\bullet}$, uniformly in time. Both families are fixed before any data are observed, so a union bound over the $\binom{d}{2}$ adjacency nulls and $d(d-1)$ direction nulls gives $\alpha_A + \alpha_D = \alpha$. Which certificates the algorithm evaluates never enters the bound, so the graph-dependent order in which pairs are examined is free. $\square$

Remark 10 (On the multiplicity layer). The union bound is the least powerful admissible way to combine e-values; e-closed testing with weighted e-Bonferroni local tests strictly dominates it

[Hartog and Lei, 2026]. It cannot be used here. Our certificates reject on the *running maximum* of an e-process, and a running maximum is a *pseudo* e-value: Ville bounds the probability that it ever crosses, but its expectation exceeds one, so it is not an e-value and closed testing does not apply. Hartog and Lei [2026] are explicit on this point, and recover a bound of $\alpha + O(\alpha^2 \log \alpha^{-1})$ only for *independent* e-processes. Every per-hypothesis e-process here is a function of the same running Gram matrix, so independence fails badly. The union bound is therefore the correct instrument in this setting rather than a lazy one.

6 Related work

Most primitives used here are standard, and it is worth saying so precisely.

Not new. That the minimum of e-values is an e-value for a union null is standard [Vovk and Wang, 2021]. Aggregating conditional-independence tests over conditioning sets into an edge-level statistic is done by PC-p [Strobl et al., 2017], which upper-bounds an edge’s p-value by the *maximum* p-value over conditioning sets—the exact p-value analogue of (3)—and controls FDR with Benjamini–Yekutieli. Adjacency as the object being tested is the premise of DAT [Amin and Wilson, 2024], which replaces the exponential set of tests with a relaxed optimisation problem solved by neural networks. One-sided, faithfulness-free, finite-sample control of false edges is exactly Theorem 2 of ϵ -CUT [Shaska and Mitra, 2025], reached by the same reasoning we use: a false-positive guarantee needs a correct null, not faithfulness. Abstention and three-valued output are not new either: FCI leaves unidentifiable edges undirected, Uehara [2026] attaches per-edge **resolved/impossible** codes with tiers that abstain when a precondition test rejects, and Prakash et al. [2024] give a bivariate direction test with an explicit inconclusive outcome. Sequential, anytime-valid e-values *inside* PC are already used by Csillag et al. [2025], who calibrate batched Fisher- z p-values into per-test e-values. Residual independence as a direction criterion is DirectLINGAM [Shimizu et al., 2011], and Ruiz et al. [2022] learn a topological ordering from likelihood-ratio scores on residuals.

What is new. Three things survive that accounting.

(i) *Error-controlled orientation beyond the Markov equivalence class.* Every method above with orientation error control is confined to the equivalence class, so an edge in no v-structure remains unorientable; every method that escapes the class decides directions by comparing scores, with no error control. Ruiz et al. [2022] is closest in statistic but proves only population-level identifiability and requires all noise terms from one scale-location family; its “sequential” refers to greedy sorting, not sequential inference. Uehara [2026] certifies *provenance*— which theorem licensed a direction—not error, and its Theorem 1 bounds expert query counts under an ideal-oracle assumption.

(ii) *A whole-graph guarantee proved rather than assumed.* Per-test anytime-validity is not new, but Csillag et al. [2025] obtain graph-level validity by assumption (their Assumption 2.5: the downstream algorithm is valid whenever its inputs are), noting the multiple-comparison concern without resolving it. Theorem 9 proves the statement over a fixed family and extends it to orientations.

(iii) *Time-uniformity on the adjacency side.* ϵ -CUT holds the same one-sided, faithfulness-free ground *at each fixed n* . Bounding the probability that a false edge is *ever* certified is a different statement, and the one a monitored run needs; Section 7.1 measures the gap. Secondly, minimising over the complete family $\mathcal{S}_k$ performs all the tests PC-p’s maximum-p bound requires, so PC-p’s zero-Type-II-error assumption is not needed.

Table 1: Probability of an unsound assertion at any point in a monitored run ($\alpha = 0.1$). “SPRINT-CD” is an anytime-valid PC variant that deletes edges against confidence-sequence certificates; both it and PC decide absence by failing to reject.

stratum	CERT-CD		SPRINT-CD	PC
	false edge	wrong arrow	lost edge	lost edge
δ -strong faithfulness holds ($n = 40$)	0.000	0.000	0.000	0.000
δ -strong faithfulness fails ($n = 20$)	0.050	0.000	0.600	0.800

7 Experiments

All experiments stream data in batches and inspect the estimate after every batch, scoring the worst case over the whole trajectory rather than at a single sample size. Code and figures accompany the paper.

7.1 What monitoring costs a fixed-sample test

Data are generated under a null conditional independence and the analyst inspects the evidence as observations accumulate. Over 2000 replications monitored to $n = 1000$, at nominal $\alpha = 0.05$ the Fisher- z test rejects in 34.6% of runs when monitored, against 5.6% evaluated once at the final sample size; the e-process stays at 2.4%. This is the gap that separates a fixed- n guarantee from a time-uniform one.

The e-process pays for this in power. Against a fixed-sample Fisher- z test at the same n , evaluated once, it detects a coefficient of 0.08 in 32% of runs against 72%, and 0.10 in 58% against 91%; the gap closes by 0.15 (96% against 100%). When the sample size can honestly be fixed in advance, a fixed-sample test is the better instrument.

7.2 Stratifying by the premise

The competing guarantee for delete-on-non-rejection requires δ -strong faithfulness. This is a demanding condition: at $\delta = 0.15$ it holds in 65% of random 6-variable DAGs and in only 19% of 7-variable graphs with one latent variable marginalised out, where the median smallest partial correlation over true adjacencies is about 0.03. Since removing an edge whose association genuinely lies inside the equivalence region is correct behaviour by construction, pooling both kinds of instance measures how often random graphs are near-unfaithful rather than whether a procedure is calibrated. We therefore stratify. Because the margin is a population quantity it can be computed before simulating data, making it cheap to rejection-sample a well-powered sample of each stratum.

7.3 Certificate-only against delete-on-non-rejection

Table 1 reports the probability of ever making an unsound assertion over a monitored run, at $\alpha = 0.1$, $d = 5$, Laplace noise.

Where the premise fails—the regime that breaks delete-on-non-rejection— CERT-CD stays inside its budget while the two comparators fail in 60% and 80% of runs. Near-unfaithful instances cost CERT-CD power: the affected pairs stay undecided rather than being asserted wrongly.

On premise-satisfying instances at $n = 3000$, every true edge is certified and 96% are also oriented, against a *Markov-equivalence-class orientation ceiling* of 33%—the fraction a CPDAG can orient at all on these graphs. That gap is the practical content of the direction certificate.

The same stratification applied to the skeleton alone ($d = 6, 120$ premise-satisfying instances) gives a probability of 0.008 that a true edge is ever lost, against 0.133 for PC re-run at each sample size on the same instances.

7.4 Latent confounders

An anytime-valid FCI variant using the same machinery recovers the oracle PAG under one marginalised latent variable: on premise-satisfying instances ($n = 80$) no oracle adjacency was ever lost at $\alpha = 0.1$, structural Hamming distance to the oracle PAG fell from 12.7 to 0.35 by $n = 4000$, and the bi-directed edge marking the latent was recovered in 7 of 7 cases where one was present.

8 Discussion

What the output does not say. The absence of an edge means undecided. CERT-CD cannot report that an edge is missing, and a pair may remain undecided indefinitely if adjacency-faithfulness fails for it—which is exactly the situation in which every delete-on-non-rejection method returns a confident wrong answer. This is the price of the rule, and we regard reporting it as preferable to concealing it.

Limitations. The direction certificate assumes causal sufficiency for the pair and a parametric noise family; Remark 8 reports what misspecification does empirically, but validity outside the family is not established. The FCI variant implements orientation rules R1–R4 and omits R5–R10, so its PAG is sound but not maximally informative. The direction process refits per pair per batch and has been evaluated only up to $d = 7$; scaling is unproven. Finally, the orientation results use Laplace noise at $d = 5$, and the misspecification study of Remark 8 is bivariate rather than graph-level.

Terminology. “Anytime FCI” [Spirtes, 2001] denotes a *computational* property—the outer loop over conditioning-set sizes may be interrupted and still return a sound graph. “Anytime-valid” here is an *inferential* property: time-uniform error control at data-dependent stopping times. The two are unrelated.

Data and code availability

Implementation, experiments and figures are available at <https://github.com/vinhqdang/SPRINT-CD>.

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